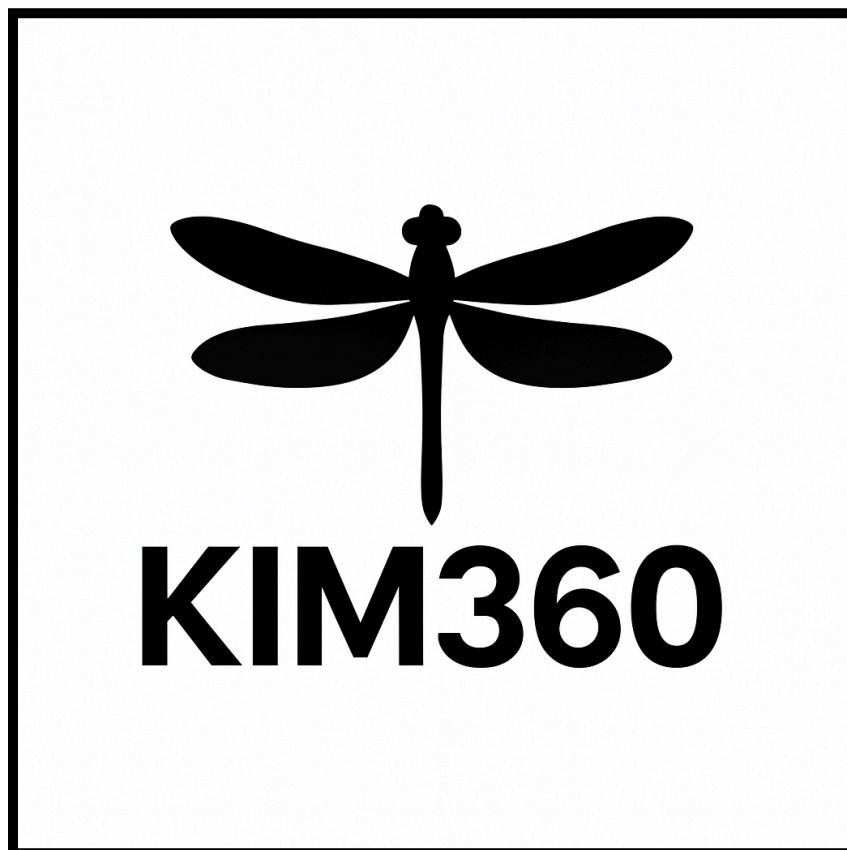


KIMinder360 Requirements Document



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Introduction

Context

Modern society faces a growing need for heightened situational awareness. In densely populated urban and suburban environments, issues such as distracted behavior, social disconnection, and unpreparedness for emergencies have become increasingly evident. According to recent public safety and behavioral studies, more than 80% of people admit to being unaware of their surroundings when using their phones or commuting, these are conditions that can lead to increased vulnerability in crises and reduced day-to-day attentiveness.

At the same time, the global wellness and self-improvement industry has seen explosive growth in mobile applications that target mindfulness, physical health, and productivity. However, the vast majority of these apps encourage passive reflection rather than active awareness, focusing on meditation timers or workout tracking instead of environmental readiness and observation skills. A tool that could repurpose this same technology to train users in real-world awareness, resilience, and preparedness would not only fill a neglected market niche but also have meaningful social benefit.

Sponsor Background

Our project sponsor, Morgan Boatman, is a video producer, author, and the owner of Winter Communications, LLC, a company specializing in creative storytelling, training, and experiential team-building. One of his signature programs, Missions and Madness, blends outdoor exploration with collaborative challenges designed to build confidence, communication, and adaptability in unpredictable settings. Through years of running these workshops, Morgan observed how simple, well-timed situational drills could dramatically improve participants' alertness and sense of agency.

To extend this training beyond in-person events, Morgan developed a catalog of short, scenario-based exercises called directives, practical prompts such as "Identify three exits in this building" or "Describe the last car that passed you." Each directive is designed to take only a short amount of time but reinforce constant environmental awareness. The challenge he faces is distributing these exercises consistently and scaling them to a larger audience without manual coordination.

The KIMinder360 project aims to transform this manual, instructor-driven system into a self-sustaining digital platform that delivers these same experiences through a smartphone app, bringing the discipline of situational awareness training to anyone, anywhere.

Problem Statement

Current Workflow

In the past, Morgan's situational awareness training was almost entirely manually done. Participants enroll in small groups, and Morgan personally communicates each directive to them via email, text, or in-person sessions. Feedback on completed tasks is shared informally through messages or follow-up discussions. There is no centralized system to schedule drills, randomize delivery, or track long-term progress.

A simplified version of the existing workflow looks like this:

1. **Directive Creation:** Morgan writes new situational drills manually.
2. **User Contact:** Directives are distributed to participants via email or text message.
3. **Completion Tracking:** Participants self-report completion informally.
4. **Analysis:** Morgan compiles results and engagement data manually if at all.

This workflow limits growth because each participant's experience depends on Morgan's direct involvement. As the number of users increases, so does the administrative burden, making scalability practically impossible.

Identified Deficiencies

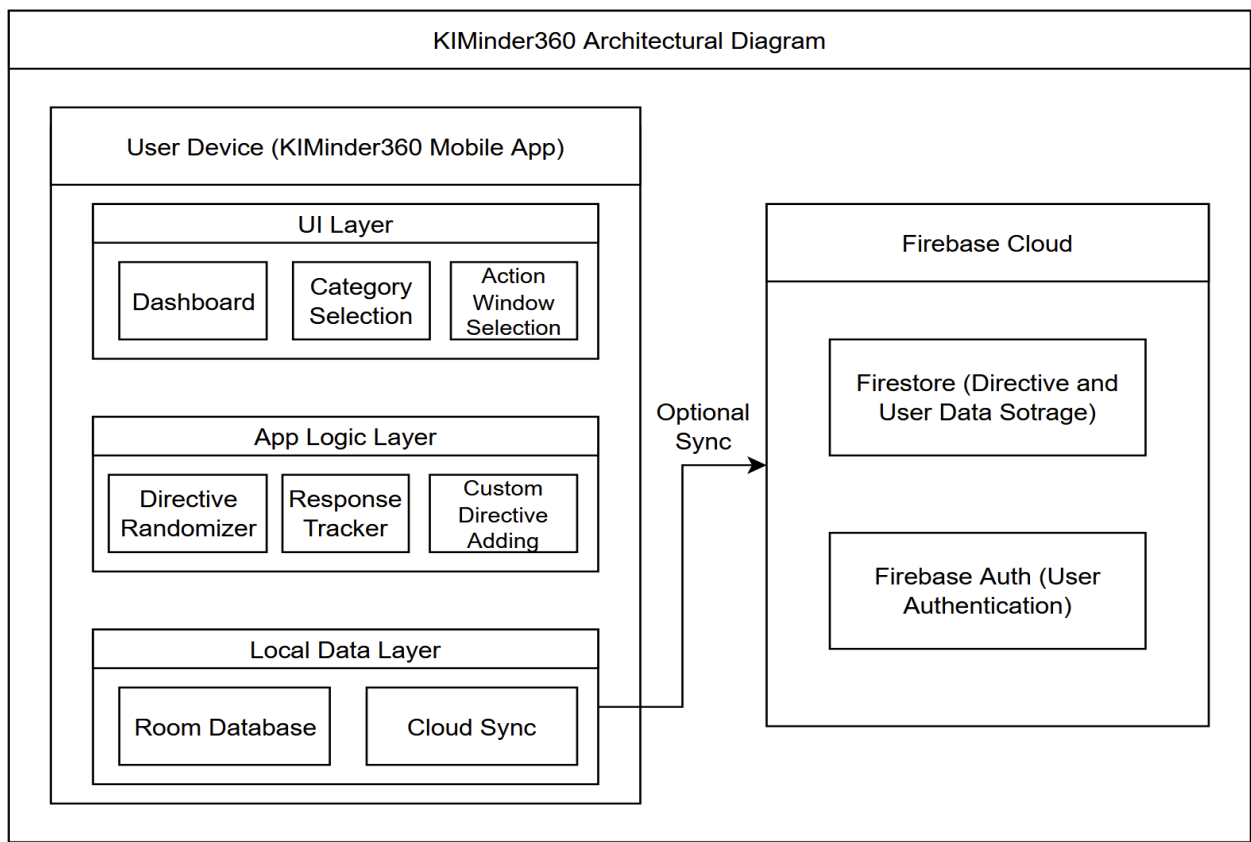
The following issues summarize the breakdowns in the current system:

- **Manual Distribution:** Every directive must be individually sent, consuming significant time.
- **No Automation:** There is no system for randomizing or scheduling directive delivery.
- **Limited Engagement:** Without gamified feedback or reminders, user participation quickly declines.
- **No Data Tracking:** There is no reliable record of user progress or directive completion rates.
- **Restricted Accessibility:** Training depends on being part of a small, local group directly supervised by Morgan.
- **Inefficient Scaling:** Each new participant increases workload linearly, preventing business growth.

These inefficiencies not only limit Morgan’s ability to expand his training system but also prevent the general public from accessing a valuable learning experience that could improve safety and situational readiness.

Solution Vision

The KIMinder360 mobile application will directly address these challenges by automating directive delivery, recording progress, and providing options and results analytics through an intuitive, gamified interface. Rather than relying on manual communication, users will define their own “Action Windows” which are periods of time during the day which they are open to receive directives. The system will then deliver randomized directives within those windows using push notifications, allowing the experience to feel spontaneous yet controlled.



[Figure 1: Architecture Diagram showing the general layout for the KIMinder360 App]

The base level of our KIMinder360 app is the local datalayer where the database of directives exists along with user preferences. This is where all persistent data is stored

that must be retained between the user closing and reopening the app. The data layer communicates with the cloud storage layer to provide optional sync capabilities with the online community directives. The firebase cloud layer is where user submitted directives will be sent and stored for other users to access them as well as where leaderboard rankings and other online features will be held. Back in the app itself, right above the local data layer is the app logic layer where all of our code accesses and manipulates the data in the databases. Functions to pick random directives, schedule them for sending, and prepare the notifications exist here. This layer also handles user input, when the user selects what categories they want to receive directives from as well as customize their active hours, functions in the app logic layer will carry out those changes. Finally, above the app logic layer is the UI layer. This is where the user interacts with the app and can make selections to customize their user preferences, view statistics, and responses to previously sent out directives. When the user makes a change to any preferences, a signal will be sent to the app logic layer to have a function execute the requested change.

Core Features

- **User Customization:** Users can select preferred directive categories (preparedness, fitness, navigation, self-awareness) and schedule their Action Windows to times when they would be available.
- **Automated Directive Delivery:** Randomized push notifications provide real-world challenges at unpredictable intervals, promoting true situational awareness.
- **Progress Tracking Dashboard:** A visual interface summarizes completed tasks, user preferences, and other relevant data.
- **Secure Local Storage:** All data including all directives, preferences, progress, and directive history is stored securely on the device allowing for total offline usage.
- **Custom Directive Creation:** Users can create and submit their own directives to their local database to incorporate them in their daily routines and action windows. This allows for total personalization of the skills you improve so that they are unique to your life.

System Overview

KIMinder360 will use a local Room database to store directive content and user data, combined with Android's notification system for timed or random prompts. Optional integration with Firebase will allow cloud synchronization with a community directive database.

Impact and Value

By transforming situational awareness into a daily micro-training experience, KIMinder360 encourages continuous practice rather than sporadic instruction. The app:

1. **Expands Reach:** Converts a small-scale coaching model into a scalable global tool.
2. **Enhances Engagement:** Uses gamification to motivate daily participation.
3. **Improves Safety and Readiness:** Builds life skills valuable in emergencies and everyday contexts.
4. **Supports Future Growth:** Provides a foundation for premium tiers, shared community challenges, or instructor dashboards.

In summary, the KIMinder360 mobile app will repurpose everyday mobile technology to allow anyone to learn valuable preparedness skills applicable in both daily life and emergencies. Its mobile app format dramatically expands reach and accessibility, turning self-improvement into an accessible, fun, and engaging experience that anyone can participate in. This will empower our sponsor, Morgan Boatman, to scale his unique training philosophy far beyond local events.

Project Requirements

Domain Level Requirements

At a high level, the KIMinder360 application must serve as a self-guided, mobile-based awareness training platform that transforms real world situational awareness tasks into interactive mini-challenges known as “directives”. The following domain level requirements define the essential system capabilities from a user perspective. Each domain level requirement is then elaborated through specific functional, performance, and environmental requirements in the subsequent sections.

1. **Automated Situational Awareness Training:** The app must provide users with automated, randomized directives to strengthen situational awareness and readiness through push notifications within user defined time windows.
2. **User Customization and Scheduling:** Users must have the ability to control when and how often directives are received through customizable Action Windows and frequency settings.

3. **Directive Management and Data Handling:** KIMinder360 must maintain a secure, local database of directives that supports storage, retrieval, categorization, and user generated submissions.
 4. **Safety, Privacy, and Legal Compliance:** The app must include a legal disclaimer and follow best practices for safety and moderation to protect users from risk and liability while maintaining offline usability.
 5. **Scalability and Future Growth:** The system should be designed with a modular architecture to allow future integration with cloud databases, monetization features, and community submissions without disrupting the core offline functionality.
-

Functional Requirements

The Functional Requirements define the specific behaviors and functions that KIMinder360 must provide to fulfill its domain level objectives. They are grouped and prioritized using the Must, Could, Should, and Won't as categories of importance.

Must Have Requirements

These features are essential to the operation of the initial release. The MVP cannot exist without them.

1. Push Notifications

User Story: As a user, I want to receive spontaneous awareness directives via push notifications during my selected Action Windows so that I can practice situational awareness throughout my day.

Priority: Must

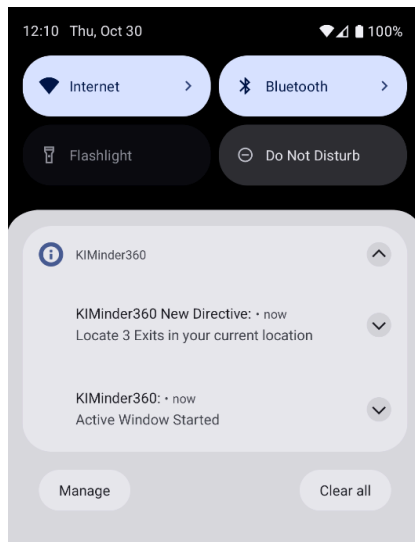
Estimate: 3 Story Points

Acceptance Criteria:

- The app must request permission to send and schedule notifications.
- Notifications must respect user-defined Action Windows and frequency limits.
- A function must exist that takes a send time, notification title, and notification content.
- Users must be able to tap a notification to view, complete, or dismiss a directive.

Description: This feature implements the primary delivery mechanism for KIMinder360's directives, ensuring that prompts are non-intrusive but consistent. Notifications use Android's AlarmManager or WorkManager for scheduling. This connects to our domain level requirement of automated situational awareness training by enabling the core functionality of sending out directives to users.

Prototype:



2. Database of Directives

User Story: As a user, I want the app to have a large and organized library of directives available offline so that I can experience new and diverse training challenges from anywhere in the world.

Priority: Must

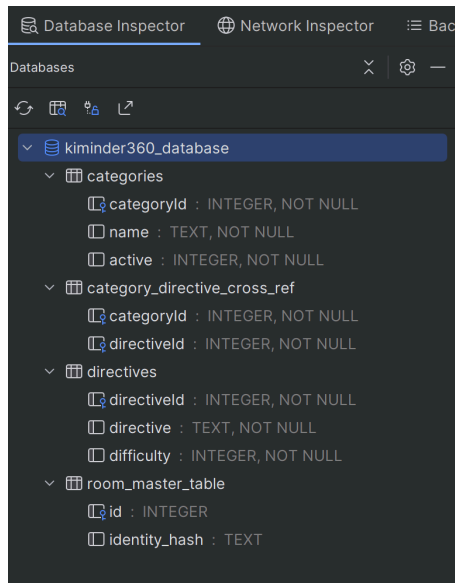
Estimate: 4 Story Points

Acceptance Criteria:

- The database must store each directive's text, categories, difficulty, and send/completion date for randomization.
- The app must be able to retrieve directives from the database to send as notifications.
- When a user completes or skips a directive, the database must update the corresponding record.
- The database must persist data across sessions and be fully available offline.

Description: This feature provides the backbone for directive delivery and tracking. It ensures that all reminders and user interactions pull from a single source. The database works in tandem with push notifications, forming the core of KIMinder360's directive system.

Prototype:



3. Category Selection

User Story: As a user, I want to be able to control the kinds of directives I get so that I can have a more personalized and enjoyable experience on the app.

Priority: Must

Estimate: 2 Story Points

Acceptance criteria:

- Must have multiple different category selection options.
- Must be able to change categories in a user-friendly way.
- Must have categories that provide more depth to the experience.

Description: This feature enhances the gamification of the app while also creating new ways to become situationally aware. This feature will provide different categories of directives in order to give the user an experience that fits their goals better. This connects to our domain level requirement of user customization and scheduling by allowing users to select only the directive categories that apply to them for the best possible experience custom to their needs.

Prototype:

4. Active Hours Select

User Story: As a user, I want to be able to select active hours throughout the week for directives to be delivered so that I can minimize interruptions and have a balance with commitments.

Priority: Must

Estimate: 3 Story Points

Acceptance Criteria:

- The feature must be capable of re-editing the hours selection.
- Must be able to show options for all days of the week.

Description: This feature allows users to customize specific hours during which they wish to receive directives or notifications. The system displays a UI that shows all seven days of the week, with each day allowing the user to select an active time range. This feature helps users lessen interruptions outside their preferred hours while having a balanced schedule. This connects to our domain level requirement of user customization and scheduling by letting the user select when they are assigned directives to work around their busy schedules.

Prototype:

< Back **Set Active Hours**

Monday ☒ Start: 09:00 End: 17:00

Tuesday ☐

Wednesday ☐

Thursday ☐

Friday ☐

Saturday ☐

Sunday ☐

Save Active Hours

Should Have Requirements

These features add significant value and usability but are not critical for the MVP launch.

1. Personal Directive Submission

User Story: As a user, I want to create my own directives so that I can personalize my experience and focus on training goals that are meaningful to me.

Priority: Should

Estimate: 4 Story Points

Acceptance Criteria:

- Users must be able to create, edit, and delete their own directives.
- Custom directives should include the text description, categories, and optional difficulty levels.
- User-created directives must remain stored locally on the device by default.
- The interface for adding directives should be simple and consistent with the existing directive system.

Description: This feature allows users to add personal directives that fit their individual goals or routines. By supporting personal directive creation, users can continue to find value in KIMinder360 even if official development or new directive sets are no longer released. This ensures that the app remains useful and relevant over time, empowering users to sustain their own situational awareness practice indefinitely. This connects to our domain level requirement of user customization by letting users create their own directives that are specific to their situations to create an experience that is best for them. It also ties into our data handling and directive management domain requirement by necessitating that our data structures can handle user created content.

2. User Engagement Selection

User Story: As someone with a schedule that fluctuates a lot, I want to be able to change how much I need to interact with directives depending on if I have a busy day or a relaxed day.

Priority: Should

Estimate: 8 User Points

Acceptance Criteria:

- From the settings menu, a selector with 3 distinct levels must be available to the user.
- The levels go from: No Engagement, Minimal Yes/No, and Full Response
- No engagement allows the user to see the notification and not interact with it at all, Minimal lets the user respond Yes or No to "Did you complete this directive", and full response lets the user type an answer to the directive.
- The effects of this selection are apparent when the user clicks on a directive notification, it will open the app and take them to a screen where they enter a response according to their selected engagement level.

Description: This feature allows the user to select 1 of three engagement levels for interacting with directives going from no engagement to full written response. This allows users to approach the situational awareness training with however much effort and attentiveness that they desire. This connects to our domain level requirement of

user customization by letting users decide how much they interact with the directives they are given to match them where they are in their situational awareness journey.

3. Difficulty Selection

User Story: As a user, I want to be able to change the difficulty level of my directives so that I can have an experience that lets me grow my skills and challenge myself.

Priority: Should

Estimate: 4 story points

Acceptance Criteria:

- There should be 3 levels of difficulty (easy, normal, hard)
- There should be a user-friendly interface to change the difficulty level.
- There should be different difficulty versions of each directive, for each level.

Description: This feature allows users to be able to change the difficulty level of the directives they receive within their category. This creates more depth to the app and a way for users to challenge themselves if that is what they're looking for. This connects to our domain level requirement of user customization by allowing our users to choose how intense the directives they receive are.

4. Skip or Delay/Postpone/Bookmark Directive Option

User Story: As a user, I want to be able to skip, delay, or bookmark a directive so that I can manage my tasks according to my schedule and complete them at a more convenient time.

Priority: Should

Estimate: 2 story points

Acceptance Criteria:

- Users must be able to choose to skip, delay, or bookmark a directive.
- Postponed directives should include an option to set a new reminder or time.
- Bookmarked directives should be saved in a list or marked for viewing later.
- Skipped directives should no longer be shown as "active".
- Users should be able to choose whether to re-activate or review skipped/bookmarked directives.

Description: This feature allows users better control for when and how they interact with directives. That way, by offering these options, users can manage their focus and prioritize their tasks. When a user chooses to skip a directive, that directive is archived. When they choose to postpone, the directive is rescheduled for a later time. When they bookmark, the directive is saved for later review. This connects to our domain level requirement of user customization by letting them postpone directives they are not able to complete at the current time and skip directives they have no interest in.

Could Have Requirements

These features are not required for initial release but may be considered for future updates or premium versions.

1. Location Functionality

User Story: As someone who travels a lot, I want directives to adapt to my current location so that I can connect with other users around me and explore the new locations I visit.

Priority: Could

Estimate: 10 Story Points

Acceptance Criteria:

- App asks permission to use Android Location Services
- Directives are tagged with locational context and relevance
- Directives ask users to meet at points of interest around them to connect with each other.
- Users can toggle on and off location based prompts

Description: Location services expand the capabilities of the KIMinder360 app to allow for more relevant directives that adapt to context and for us to connect users with each other through the app. Implementation is optional in the MVP. This connects to our domain level requirement of automated situational awareness training by encouraging training that is more in depth by connecting users with each other or having them do specific tasks that could only be done in their current location.

2. Multipart Directives

User Story: As someone who plays a lot of games, I want directives that span multiple parts or steps so that I can engage in progressive challenges similar to quests in games.

Priority: Could

Estimate: 4 Story Points

Acceptance Criteria:

- Each multipart directive comes in a set that the user chooses to start.
- These sets track user progress and completion per step so they can start and stop as they please.
- UI clearly indicates progression and how many steps are left. (X/Y or 4/5)

Description: This requirement adds the feature of specialized multipart directives that build on one another with a set theme. Through this feature we can improve user retention and enjoyment through gamified mechanics while also implementing

advanced training that single directives won't allow for. This connects to our domain level requirement of automated situational awareness training by pushing our users to improve their skills even more through chains of directives that build on one another.

3. Completed Directives List

User Story: As a user, I want to view a list of directives I have completed so that I can track my progress and reflect on my situational awareness practice.

Priority: Could

Estimate: 3 Story Points

Acceptance Criteria:

- The list must display the directive, completion status, and completion date.
- The list should update automatically when new directives are assigned (sent to the user) or completed.
- The user should be able to scroll through the list or filter completed directives by date or category.

Description: This feature provides users with a visual record of their progress. It allows them to see which directives they've completed and encourages consistent engagement. It would later become a component of the broader "Dashboard" that displays user progress or performance trends. This connects to our domain level requirement of directive management and data handling by forcing us to create in depth databases that keep track of user activity and directive history.

4. Monetization

User Story: As a user, I want the option to purchase new directive categories or packs so that I can expand my experience and explore different types of situational awareness challenges.

Priority: Could

Estimate: 2 Story Points

Acceptance Criteria:

- The app could offer purchasable directive packs or categories as optional upgrades.
- Purchased content must unlock immediately after a successful transaction.
- The system must prevent unauthorized access to premium content.
- Monetization options must not interfere with the core free experience.

Description: This feature enables future revenue generation by allowing users to buy new directive categories, packs, or additional features such as themes or extra cloud storage for sharing directives online. Monetization would be handled securely through the respective app store's purchase system(s) and integrated within the user interface

as optional enhancements. This connects to our domain level requirement of Scalability and growth by giving us the option to develop new content in the form of directive packages and categories that the user can purchase.

5. Leaderboards

User Story: As a user, I want to view and compare my directive completion progress on a leaderboard so that I can be engaged and motivated to complete my directives

Priority: Could

Estimate: 4 Story Points

Acceptance Criteria:

- The leaderboard could include mutual connections and/or household members.
- The top positions of the leaderboard could display the top ten users based on the number of completed directives.
- Each entry could include the user's name and the total completed directives
- The user's own positions could be (optionally) displayed below the leaderboard for motivation.

Description: This feature allows users to compete in completing directives and be displayed in a leaderboard among peers or household members. The leaderboard would display the total number of directives that the top users have completed. This connects to our domain level requirement of automated situational awareness training by implementing a gamification mechanic that will encourage the user to try harder directives and complete more in any given period of time.

6. Cloud Database Shared Directives

User Story: As a user, I want my directives to be stored and shared through a cloud database so that my data is accessible across devices and can be shared with other users when needed.

Priority: Must

Estimate: 5 Story Points

Acceptance Criteria:

- Users must be able to access their directives across multiple devices with synchronization.
- The system must support shared directives between approved users.
- Directives must be stored in a centralized cloud database.

Description: This feature provides centralized data management for all information with directives. By using a secure cloud database, directives are accessible and synced across different devices and accounts. This system allows for shared access among users, giving them control to collaborate with chosen users. This connects to our

domain level requirement of scalability and growth by letting our community think of directives that our development team could not and contributing them to the community database where our content will grow naturally.

The requirements above collectively ensure that KIMinder360 meets both sponsor and user expectations for functionality, engagement, and safety.

- **Must-Have Features** form the MVP baseline ensuring the app performs its core awareness-training role.
 - **Should-Have Features** enhance user autonomy and motivation.
 - **Could-Have Features** define future pathways for social engagement and scalability.
 - **Won't-Have Features** clarify project boundaries to maintain focus on deliverables achievable within the initial development timeframe (None have been outlined at this time).
-

Performance Requirements

The KIMinder360 system is designed to provide a seamless and responsive user experience across a wide range of Android devices, with particular emphasis on timing accuracy, usability, and offline reliability. These non-functional requirements ensure that the system performs efficiently under normal conditions and maintains usability standards appropriate for a mobile situational awareness training tool. Below we will detail specific and measurable requirements that are verifiable through testing or inspection.

1. System Responsiveness

- **App Launch Time:** The application should launch from a cold start within 4 seconds on devices meeting minimum specifications of Android 10+ and at least 4 GB RAM. (Verification: stopwatch measurement or Android profiler.)
- **Directive Load Time:** Directives retrieved from the local Room database should appear in the user interface within 1 second of request. (Verification: test using multiple database sizes up to 10,000 directives.)
- **Domain Requirement Connection:** System responsiveness ties into our domain requirement of user customization and scheduling because the app loading quickly means that users can interact with a snappy UI to customize their preferences.

2. Data Management and Storage

- **Offline Functionality:** All directive and user data shall remain fully accessible and functional offline with no reliance on network connectivity for core features. (Verification: offline testing with Wi-Fi and data disabled.)
- **Data Integrity:** The system shall ensure zero data loss during app restarts, power loss, or when switching between online/offline states. (Verification: database stress testing and power-failure simulations.)
- **Domain Requirement Connection:** Data management and storage connects to Directive Management and Data handling by ensuring that we have a secure and persistent collection of databases that contain all user preferences and directives in a way that makes CRUD operations easy. It also ties in to safety and privacy by ensuring that all data is held securely and locally on the phone where it will not be shared with anyone.

3. Usability and User Learning

- **Ease of Learning:** A first-time user should be able to configure Action Windows and select directive categories within 3 minutes of initial launch without external instruction. (Verification: usability testing with 10 novice users.)
- **Task Completion Time:** Advanced users should be able to log directive completion or add a custom directive in under 10 seconds from the main dashboard. (Verification: time-trial usability test.)
- **User Satisfaction Threshold:** During pilot testing, at least 80% of participants should report that app navigation and notification timing feel “intuitive” or “non-intrusive.” (Verification: post-test user survey.)
- **Domain Requirement Connection:** Having the app be easy to use and learn connects to our domain level requirement of user customization and scheduling because we want users to be able to pick up the app and intuitively use it within minutes so that they can easily start customizing their preferences.

4. Performance Efficiency

- **CPU and Memory Use:** The application will consistently use under 10% CPU utilization and under 150 MB memory usage during background operation on a mid-range Android device. (Verification: profiling via Android Studio.)
- **Battery Consumption:** Background services (notifications, timers, and database sync) shall consume less than 3% battery per hour of active use. (Verification: monitored via Android battery usage statistics.)
- **Domain Requirement Connection:** Performance requirements relate to the overall goal of automated situational awareness training by making sure that the

user notices the app as little as possible. When KIMinder360 is running in the background with minimal performance cost, all the user is aware of is the convenient method of situational awareness training where the app handles everything without issue.

Environmental Requirements

The KIMinder360 mobile application operates within several environmental constraints defined by hardware capabilities, software frameworks, and sponsor specifications. These requirements ensure compatibility, compliance, and maintainability of the final system.

1. Hardware and Platform Constraints

- **Target Platform:** KIMinder360 will run on Android smartphones and tablets running Android 10 (API Level 29) or higher. (Rationale: Ensures compatibility with current Android notification APIs and Room database libraries.)
- **Required Permissions:** KIMinder360 may access location services and will need permission to send notifications.

2. Software and Development Constraints

- **Programming Language:** All development will be done in Kotlin using Android Studio as our development environment of choice. (Rationale: Aligns with modern Android development standards and the sponsor's platform goals.)
- **Database System:** KIMinder360 will use Room Persistence Library for local data management to ensure offline capability, data security, and type safety.
- **Notification System:** The app will utilize Android NotificationManager APIs for push alerts and directive delivery.
- **Optional Cloud Integration:** Future versions may employ Firebase Realtime Database or Firestore for community directive sharing, but the MVP (minimum viable product) must remain fully functional offline.

3. External Standards

- **Ethical Directives:** The KIMinder360 system will not generate or encourage directives that violate laws, safety protocols, or public property restrictions.

- **Legal Consideration:** A liability disclaimer will be displayed to the user during the onboarding process and in the settings menu to ensure the user is aware of personal responsibility for their actions.

4. Operational Environment

- **Usage Context:** The system is designed to be used in all environments (indoors and outdoors) including areas with limited or no internet connectivity.
 - Example Scenario: A user walking through a city receives a directive such as “Identify two exits in the next building you enter,” completes it offline, and logs completion.
 - **Deployment Environment:**
 - Distributed via Google Play Store for Android devices.
 - Testing environments shall include both emulated and physical devices across different manufacturers.
-

Potential Risks

No software system can fully prevent misuse or misinterpretation once it is deployed to the public, particularly one that intentionally delivers unpredictable real-world challenges. Because KIMinder360 encourages users to act in physical environments, risk management must go beyond technical failure and include behavioral and safety considerations. The following section outlines the most significant risks relevant to the system’s design and operation, their potential impact on users and the sponsor’s brand, and planned mitigations.

1. Dangerous or Unreviewed Community-Submitted Directives

Description:

A core goal of KIMinder360 is to cultivate an engaged user community capable of contributing new situational-awareness directives. However, allowing open submission introduces the risk that unsafe, illegal, or harmful tasks could be uploaded and distributed before proper review. Examples include directives that encourage trespassing, confrontation with strangers, or physical hazards.

Impact:

If a user is injured or feels endangered after following such a directive, the result could be serious physical harm, public backlash, or legal liability for Winter Communications.

Even a single incident could damage user trust and undermine the program's credibility as a personal-development tool.

Likelihood:

Moderate. While most users will act in good faith, the presence of malicious or careless submissions is a persistent risk in any open community.

Mitigation Strategies:

- Implement a **mandatory moderation queue** where all user-submitted directives require administrator approval before being added to the community database.
- Introduce **content flagging and reporting** mechanisms so inappropriate prompts can be quickly reviewed or removed.
- Provide **clear community guidelines and disclaimers** outlining acceptable directive types and user responsibility.
- Develop **automated screening tools** using keyword filters to catch high-risk submissions early.

2. Excessive or Poorly Timed Notifications

Description:

The system's engagement model depends on random notifications delivered within user-defined "Action Windows." If notifications are too frequent, occur at inappropriate times, or fail to respect user schedules, they can become intrusive or distracting.

Impact:

Annoying notifications may cause users to disable alerts, uninstall the app, or leave negative reviews. In some contexts such as while driving or working, poorly timed directives could even create safety concerns by diverting attention at critical moments.

Likelihood:

High. Notification fatigue is a well-documented issue across all mobile platforms, especially for applications using push-based engagement.

Mitigation Strategies:

- Require users to **explicitly configure Action Windows** and maximum daily directive limits during onboarding.
- Employ **directive spacing computation** to ensure directives have adequate amounts of time between them so that the user does not feel they are occurring too frequently.

- Include **“Do Not Disturb” synchronization** to prevent alerts during meetings, driving, or quiet hours.
- Add a **frequency selection** from 1 to 10 to let the user customize how often directives are handed out.

3. Lack of Directive Randomization

Description:

Directive repetition due to insufficient randomization or limited directive pools can diminish user motivation and the perceived value of the app. When users repeatedly receive identical or overly similar tasks, engagement can decline sharply.

Impact:

The psychological effect is not directly harmful but leads to **reduced retention**, poor reviews, and a weakened training impact. Over time, this could undermine the app’s core purpose: encouraging continuous awareness through novelty.

Likelihood:

Moderate to High. Without careful randomization algorithms and a sufficiently large directive database, repetition becomes inevitable for active users.

Mitigation Strategies:

- Implement **weighted randomization** that prioritizes uncompleted or rarely used directives.
- Track recent **user history** to prevent duplicates within a rolling time window.
- Continuously **expand the directive database** through verified community submissions and sponsor updates to decrease the chance of repeated directives.
- Expand **directive categories** so repetition within the same theme feels more diverse.

4. Ambiguous or Misinterpreted Directives

Description:

Directives that are poorly phrased or lack sufficient context can lead to confusion. In extreme cases, a user might misread a task and perform an action that is socially inappropriate or physically unsafe (“Find the nearest exit” interpreted as “test the emergency door alarm”).

Impact:

Ambiguous directives pose both **safety and reputational risks**. Misinterpretation could result in user embarrassment, property damage, or interactions with law enforcement. Additionally, it can diminish user trust if directives seem unclear or misleading.

Likelihood:

Moderate. Even well-written instructions may be interpreted differently by users of varying cultural or linguistic backgrounds.

Mitigation Strategies:

- Use **clear and concise directive language** validated through user testing.
- Include **contextual examples or safety notes** where necessary (“Do not enter restricted areas”).
- Provide an **in-app feedback mechanism** for users to report unclear or unsafe directives.

Risk Summary Table

Risk	Impact Severity	Likelihood	Overall Priority
Dangerous or unreviewed directives	Very High	Moderate	Critical
Excessive or poorly timed notifications	Moderate	High	High
Lack of directive randomization	Low	Moderate/High	Moderate
Ambiguous or misinterpreted directives	High	Moderate	High

Conclusion

While none of these risks are catastrophic to system viability, they directly affect user safety, satisfaction, and brand integrity. The most severe risks being unreviewed directives and ambiguous prompts can be mitigated through strong moderation, rigorous content review, and clear communication with users. Properly managed, these risks become opportunities to strengthen KIMinder360's credibility as a responsible, engaging, and safety-conscious awareness training platform.

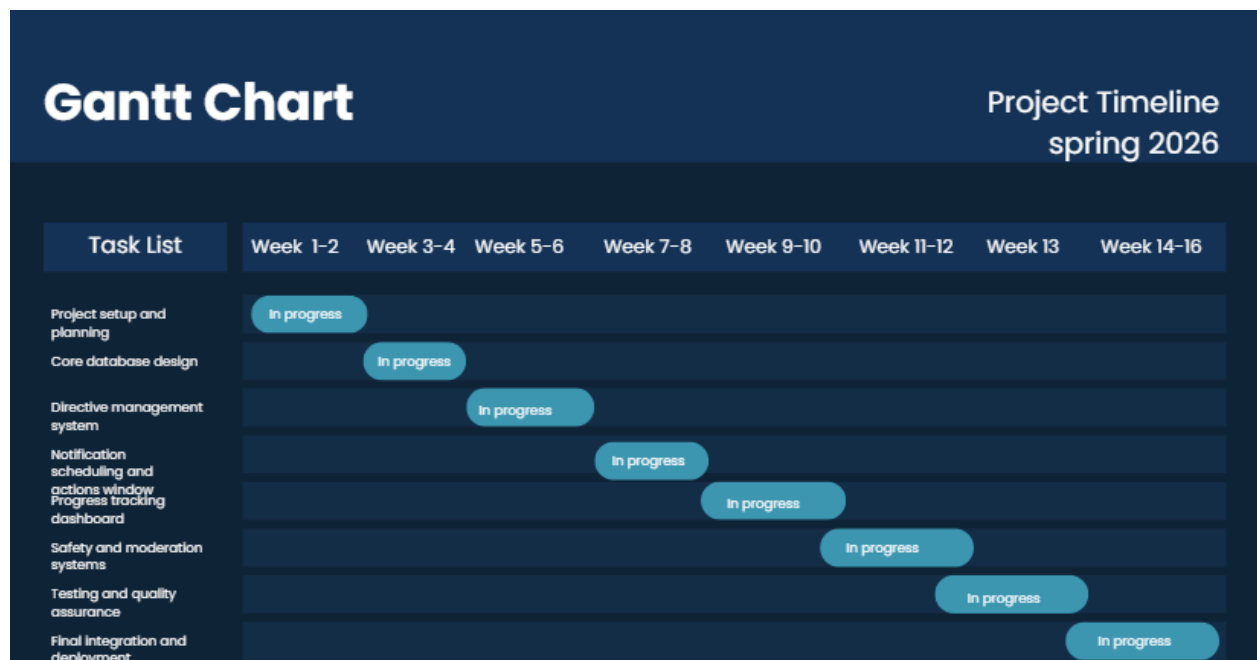
Project Plan

The development of KIMinder360 will follow a structured, milestone based plan designed to ensure timely completion, organized progression, and iterative refinement of core features. The plan spans approximately four months, aligning development tasks with the system's functional requirements such as data storage, scheduling active hours, user customization, notification delivery, and risk-mitigation features. Each milestone represents a tangible, testable output that contributes directly to the completion of a core system component.

Milestones and Timeline (Planned for Spring 2026 semester)

1. Milestone 1: Project Setup & Planning (Week 1–2)
 - Establish project repository, version control, and task tracking.
 - Define final app architecture (MVVM pattern with Room and ViewModel).
 - Confirm tech stack: Kotlin, Room Database, Android Notification Manager, and optional Firebase integration.
 - Deliverable: Development environment configured and documentation initialized.
2. Milestone 2: Core Database Design (Week 3–4)
 - Create Room database schema for directives, user preferences, and history tracking.
 - Implement data access objects (DAOs) and entity models.
 - Deliverable: Fully functional local data layer with CRUD operations.
3. Milestone 3: Directive Management System (Week 5–6)
 - Implement directive creation, editing, deletion, and retrieval features.
 - Enable user-defined categories and personalization.
 - Deliverable: Users can view, add, and modify directives through a simple UI.
4. Milestone 4: Notification Scheduling and Action Windows (Week 7–8)
 - Develop system for user-configured Action Windows.

- Integrate random directive scheduling and delivery using Android notifications.
 - Deliverable: Users receive directives at appropriate, randomized times.
5. Milestone 5: Progress Tracking Dashboard (Week 9–10)
- Design UI for displaying completed directives, progress statistics, and user preferences.
 - Include analytics features for engagement tracking.
 - Deliverable: Functional progress dashboard connected to local database.
6. Milestone 6: Safety & Moderation Systems (Week 11–12)
- Implement directive validation, flagging, and reporting tools.
 - Add filters for unsafe or ambiguous content.
 - Deliverable: Functional moderation and risk-mitigation system.
7. Milestone 7: Testing and Quality Assurance (Week 13)
- Conduct unit and integration testing of database, notifications, and UI components.
 - Evaluate usability and timing accuracy of notifications.
 - Deliverable: Stable, bug-free beta version.
8. Milestone 8: Final Integration & Deployment (Week 14–16)
- Prepare app for release including icon, splash screen, and onboarding screens.
 - Conduct final presentation and sponsor review.
 - Deliverable: Completed app ready for public release and submission.



[Figure 2: Gantt Chart showing our development plan timeline for Spring 2026]

Discussion

This execution plan ensures that each development phase builds logically on the previous one. Core features such as database operations and notifications are prioritized early to allow for the bulk of development time for their implementation to be perfected. Risk management strategies such as user directive review and notification scheduling to optimize engagement are integrated mid-project to align with safety concerns identified in the risk assessment.

By following this roadmap, we will deliver a functional, tested, and scalable awareness training application by the end of the four-month cycle. Future iterations can then build on this foundation to add cloud synchronization, gamification systems, and expanded community features.

Conclusion

Modern life rewards efficiency and digital engagement, but often at the expense of awareness, mindfulness, and readiness. The KIMinder360 project addresses this imbalance by transforming the smartphone which is typically a source of distraction into a platform for building observation, preparedness, and situational awareness skills. Rooted in Morgan Boatman's Missions and Madness training program, the system represents an innovative step in making situational awareness education accessible to anyone, anywhere.

Throughout this document, we have outlined both the importance and feasibility of the project. We identified that Morgan's current manual process of distributing directives is unsustainable at scale and prone to inconsistency. Our proposed solution is a mobile application that automates directive scheduling, randomization, and user feedback. It directly solves the limitations of Morgan's manual implementation while preserving the spontaneous nature and challenge that define the program's philosophy. By doing so, KIMinder360 not only improves the efficiency of directive delivery but also broadens its reach to a much larger audience.

This paper contributed to project progress by defining an outline of the system's features, problems being solved, and the various aspects of each requirement as well as identifying critical risks that could affect user safety and long-term adoption. We analyzed potential hazards such as unreviewed community submissions, poor notification timing, lack of directive randomization, and ambiguous wording. These risks informed key design safeguards, including moderation, careful scheduling, and clearer communication standards which are insights that will guide our development moving forward.

In summary, KIMinder360 is more than a mobile app, it is a platform for reshaping how people interact with their surroundings. The groundwork established here demonstrates a clear understanding of the challenges, risks, and user needs central to delivering a responsible and impactful product. As development continues, the team remains confident that KIMinder360 will achieve its mission: to make everyday situational awareness a habit, not an effort, and to empower users to engage with the world more attentively, safely, and confidently.